CLEANCITY SERVICES

# Penetration Test and Hardening of Scheduling Server

## Task 1 – Lab Setup & Initial Reconnaissance

| Document | Practical Evidence & Technical Report |
|---|---|
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| Client / Project | CleanCity Services |
| Target | Ubuntu 22.04 VM – 172.16.229.131 |
| Application | Damn Vulnerable Web Application (DVWA) |
| Assessment tools | Nmap 7.80, Gobuster 2.0.1, curl |
| Assessment date | 12 November 2026 (as recorded in evidence) |
| Environment | Isolated local lab VM (VMware) |

### Prepared for supervisor review

This report documents the commands executed, their purpose, observed output, and supporting screenshots. It is intended to be understandable independently of the GitHub repository.

# Document Guide

This practical report follows the actual sequence shown in the supplied terminal and browser evidence: preparing Ubuntu, installing the DVWA stack, configuring the database, verifying the application, identifying the VM address, running Nmap, and enumerating DVWA paths with Gobuster.


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# 1. Scope and Objective

The objective of Task 1 was to create an Ubuntu 22.04 target VM, install DVWA and its dependencies, identify exposed network services with Nmap, enumerate web paths with Gobuster or DIRB, and preserve the commands and results as reproducible evidence.

Authorized scope: All activity documented here was performed against the intentionally vulnerable DVWA installation in the user's isolated lab VM. The target address was 172.16.229.131. Do not expose DVWA to the public internet.

Lab topology: The base Ubuntu operating system stores the project repository and report files. A separate Ubuntu 22.04 VM (hostname ubunto, account shown as intern) hosts Apache, MariaDB, PHP and DVWA. Commands targeting localhost were run inside that VM.

# 2. Ubuntu Preparation and Service Installation

## 2.1 Update package indexes

```
sudo apt update
```

Purpose: Refreshes the local APT package index so Ubuntu can locate current package metadata from configured repositories. The screenshot shows the Ubuntu Jammy repositories and Google Chrome repository being checked.

```
intern@ubunto:~$ sudo apt update
[sudo] password for intern:
Hit:1 http://pk.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu jammy-updates InRelease
Get:4 https://dl.google.com/linux/chrome-stable/deb stable InRelease [2,548 B]
Hit:5 http://pk.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:6 https://dl.google.com/linux/chrome-stable/deb stable/main amd64 Packages [1,40
Fetched 3,956 B in 3s (1,530 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
473 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

*Figure 1. Updating Ubuntu package indexes.*

## 2.2 Install Apache, MariaDB, PHP and supporting packages

```
sudo apt install apache2 mariadb-server php libapache2-mod-php php-mysql php-gd php-curl php-mbstring
php-xml php-zip git unzip -y
```

Purpose: Installs the web server, database server, PHP runtime and PHP extensions required by DVWA, along with Git and unzip utilities. The package installation output confirms the requested packages were selected and configured.

```
intern@ubunto:~$ sudo apt install apache2 mariadb-server php libapache2-mod-php php-mysql php-gd php-cur
l php-mbstring php-xml php-zip git unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-26ubuntu3.2).
unzip set to manually installed.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils galera-4 gawk git-man libapache2-mod-php8.1 libapr1
  libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libcgi-fast-perl libcgi-pm-perl
  libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl liberror-perl libfcgi-bin libfcgi-perl
  libfcgi0ldbl libhtml-template-perl libmariadb3 libmysqlclient21 libonig5 libsigsegv2 libsnappy1v5
  libterm-readkey-perl liburing2 libzip4 mariadb-client-10.6 mariadb-client-core-10.6 mariadb-common
```

*Figure 2. Installing DVWA dependencies.*

## 2.3 Enable and start services

```
sudo systemctl enable apache2
sudo systemctl enable mariadb
sudo systemctl start apache2
sudo systemctl start mariadb
```

Purpose: Enables Apache and MariaDB to start automatically at boot, then starts both services in the current session.

```
intern@ubunto:~$ sudo systemctl enable apache2
[sudo] password for intern:
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
intern@ubunto:~$ sudo systemctl enable mariadb
Synchronizing state of mariadb.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mariadb
intern@ubunto:~$ sudo systemctl start apache2
intern@ubunto:~$ sudo systemctl start mariadb
intern@ubunto:~$
```

*Figure 3. Enabling and starting Apache and MariaDB.*

## 2.4 Verify service status

```
sudo systemctl status apache2
sudo systemctl status mariadb
```

Purpose: Checks whether each service is running. The evidence shows Apache 2.4.52 and MariaDB 10.6.23 in an active (running) state.

```
intern@ubunto:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:38:52 PKT; 2min 24s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 21687 (apache2)
      Tasks: 6 (limit: 4548)
     Memory: 13.1M
        CPU: 102ms
     CGroup: /system.slice/apache2.service
             ├─21687 /usr/sbin/apache2 -k start
             ├─21689 /usr/sbin/apache2 -k start
             ├─21690 /usr/sbin/apache2 -k start
             ├─21691 /usr/sbin/apache2 -k start
```

*Figure 4. Apache service status.*

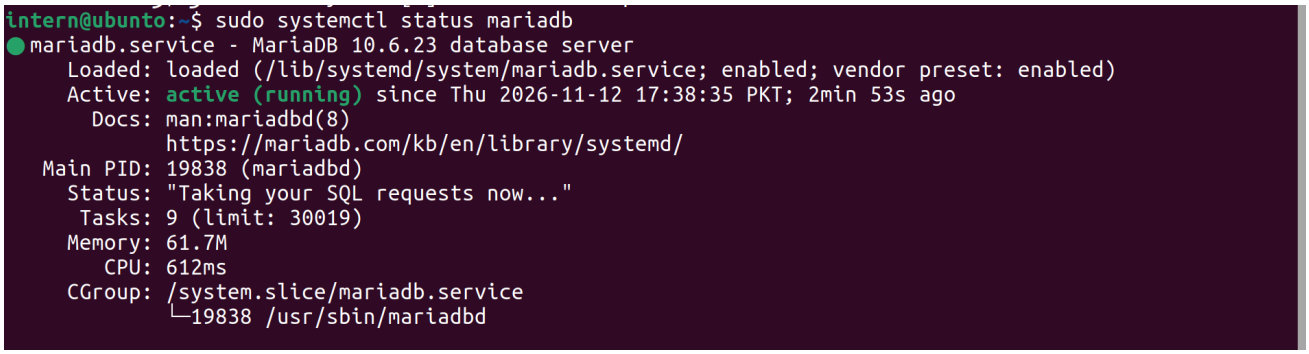



*Figure 5. MariaDB service status.*

### 2.5 Verify installed versions

```
apache2 -v
php -v
mysql --version
```

Purpose: Displays the installed server/runtime versions. The screenshots record Apache 2.4.52, PHP 8.1.2 and MariaDB 10.6.23.




*Figure 6. Version verification for Apache, PHP and MariaDB.*

## 3. DVWA Deployment and Database Configuration

### 3.1 Clone DVWA into the Apache web root

```
sudo git clone https://github.com/digininja/DVWA.git /var/www/html/DVWA
```

Purpose: Downloads the DVWA source repository into Apache's document root under the DVWA directory.

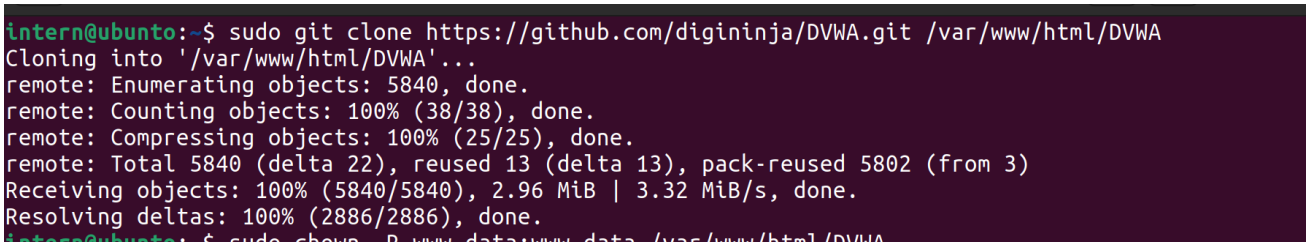



*Figure 7. Cloning the DVWA repository.*

### 3.2 Set Apache ownership and create the configuration file

```
sudo chown -R www-data:www-data /var/www/html/DVWA
cd /var/www/html/DVWA/config
sudo cp config.inc.php.dist config.inc.php
```

Purpose: Assigns the DVWA directory to the Apache service account and creates the active configuration file from the distributed template.

```
intern@ubunto:~$ sudo chown -R www-data:www-data /var/www/html/DVWA
intern@ubunto:~$ cd /var/www/html/DVWA/config

sudo cp config.inc.php.dist config.inc.php
intern@ubunto:/var/www/html/DVWA/config$ ls -l /var/www/html/DVWA/config/
total 8
-rw-r--r-- 1 root     root     2437 17:46 12  نومبر config.inc.php
-rw-r--r-- 1 www-data www-data 2437 17:46 12  نومبر config.inc.php.dist
```

*Figure 8. Setting ownership and preparing DVWA configuration.*

## 3.3 Create the DVWA database and dedicated database user

```
sudo mysql
CREATE DATABASE dvwa;
CREATE USER 'intern'@'localhost' IDENTIFIED BY '[LAB_PASSWORD]';
GRANT ALL PRIVILEGES ON dvwa.* TO 'intern'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Purpose: Creates the application database, creates a dedicated local database account, grants it privileges on the DVWA database, and reloads privilege tables. The password is intentionally masked in this report; use a unique lab-only password and never publish credentials.

```
intern@ubunto:/var/www/html/DVWA/config$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 10.6.23-MariaDB-0ubuntu0.22.04.1 Ubuntu 22.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.018 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa.* TO 'intern'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
intern@ubunto:/var/www/html/DVWA/config$
```

*Figure 9. Creating the DVWA database and granting the dedicated user access. Credential text is redacted.*

## 3.4 Configure DVWA database connection

```
$_DVWA[ 'db_server' ] = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ] = '[DB_USER]';
$_DVWA[ 'db_password' ] = '[LAB_PASSWORD]';
$_DVWA[ 'db_port' ] = '3306';
```

Purpose: Configures DVWA to connect to the local MariaDB service using the database and dedicated account created above. Credentials are masked in this report.

```
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
#   See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = getenv('DB_SERVER') ?: '127.0.0.1';

$_DVWA[ 'db_port' ]     = getenv('DB_PORT') ?: '3306';
```

*Figure 10. DVWA database connection settings. Credential values are redacted.*

### 3.5 Complete DVWA setup in the browser

The DVWA setup page is used to create/reset the application's database tables. The supplied browser evidence shows the DVWA home page after setup, confirming that the application interface is available.

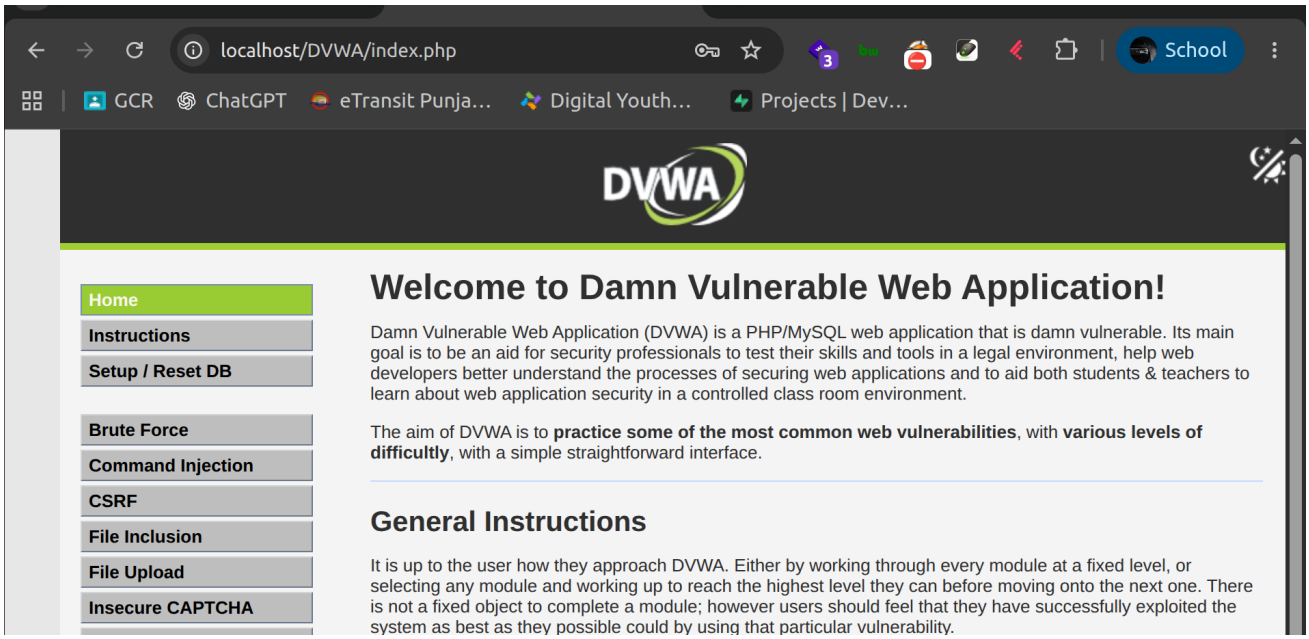



*Figure 11. DVWA application successfully opened in the browser.*

# 4. Network Identification and Nmap Reconnaissance

## 4.1 Identify the VM IPv4 address

```
ip addr
```

Purpose: Displays network interfaces and assigned IP addresses. The evidence shows interface ens33 with IPv4 address 172.16.229.131/24.

```
intern@ubunto:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d7:cb:fb brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.16.229.131/24 brd 172.16.229.255 scope global dynamic noprefixroute ens33
       valid_lft 1274sec preferred_lft 1274sec
    inet6 fe80::5465:289e:dc0b:8d38/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

*Figure 12. VM network interface and assigned IPv4 address.*

## 4.2 Install and verify Nmap

```
sudo apt install nmap -y
nmap --version
```

Purpose: Installs Nmap from Ubuntu's package repository and confirms the installed version. The screenshots record Nmap 7.80.

```
intern@ubunto:~$ sudo apt install nmap -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libblas3 liblinear4 lua-lpeg nmap-common
Suggested packages:
  liblinear-tools liblinear-dev ncat ndiff zenmap
The following NEW packages will be installed:
  libblas3 liblinear4 lua-lpeg nmap nmap-common
0 upgraded, 5 newly installed, 0 to remove and 473 not upgraded.
Need to get 5,973 kB of archives.
After this operation, 26.3 MB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu jammy/main amd64 libblas3 amd64 3.10.0-2ubuntu1 [228 kB]
```

*Figure 13. Installing Nmap.*

```
Setting up lua-lpeg:amd64 (1.0.2-1) ...
Setting up libblas3:amd64 (3.10.0-2ubuntu1) ...
update-alternatives: using /usr/lib/x86_64-linux-gnu/blas/libblas.so.3 to provide /usr/lib/x86_64-linux-
gnu/libblas.so.3 (libblas.so.3-x86_64-linux-gnu) in auto mode
Setting up nmap-common (7.91+dfsg1+really7.80+dfsg1-2ubuntu0.1) ...
Setting up liblinear4:amd64 (2.3.0+dfsg-5) ...
Setting up nmap (7.91+dfsg1+really7.80+dfsg1-2ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...

Progress: [ 95%] [#########################################################################################....]
```

*Figure 14. Package installation completion.*

```
intern@ubunto:~$ nmap --version
Nmap version 7.80 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.2 nmap-libssh2-1.8.2 libz-1.2.11 libpcre-8.39 libpcap-1.10.1 nm
p-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
intern@ubunto:~$
```

*Figure 15. Verifying the installed Nmap version.*

## 4.3 Run service/version and OS detection scan

```
sudo nmap -sV -A 172.16.229.131 -oN reconnaissance/nmap/initial-scan.txt
```

Purpose: -sV probes open ports to identify services and versions; -A enables OS detection, version detection, default scripts and traceroute; -oN saves the scan in normal text format. The scan was performed against the lab VM IP.

```
1  # Nmap 7.80 scan initiated Thu Nov 12 18:08:35 2026 as: nmap -sV -A -oN reconnaissance/nmap/
   initial-scan.txt 172.16.229.131
2  Nmap scan report for ubunto (172.16.229.131)
3  Host is up (0.000059s latency).
4  Not shown: 999 closed ports
5  PORT   STATE SERVICE VERSION
6  80/tcp open  http    Apache httpd 2.4.52 ((Ubuntu))
7  |_http-server-header: Apache/2.4.52 (Ubuntu)
8  |_http-title: Apache2 Ubuntu Default Page: It works
9  Device type: general purpose
10 Running: Linux 2.6.X
11 OS CPE: cpe:/o:linux:linux_kernel:2.6.32
12 OS details: Linux 2.6.32
13 Network Distance: 0 hops
14
15 OS and Service detection performed. Please report any incorrect results at https://nmap.org/
   submit/ .
16 # Nmap done at Thu Nov 12 18:08:43 2026 -- 1 IP address (1 host up) scanned in 8.16 seconds
```

*Figure 16. Nmap scan output saved as initial-scan.txt.*

## 4.4 Nmap results

| Item | Observed result |
|---|---|
| Target | ubunto – 172.16.229.131 |

| Item | Observed result |
|---|---|
| Host state | Up |
| Scanned ports | 1000 default TCP ports; 999 reported closed |
| Open port | 80/tcp |
| Service | HTTP |
| Detected server | Apache httpd 2.4.52 (Ubuntu) |
| HTTP title at / | Apache2 Ubuntu Default Page: It works |
| OS fingerprint | Linux 2.6.32 (Nmap estimate; not independently confirmed) |

Interpretation: Nmap identified Apache on TCP port 80. The title is for the web server root path (/). DVWA is installed under the /DVWA/ subdirectory, so the root-page title does not mean DVWA is absent. The browser and curl checks documented below show that DVWA responds under that subdirectory.

Accuracy note: Nmap's Linux 2.6.32 fingerprint is an automated estimate and should not be treated as a confirmed kernel version.

# 5. HTTP Verification with curl

## 5.1 Install curl

```
sudo apt install curl -y
```

Purpose: Installs curl, a command-line HTTP client used here to inspect server response headers.

```
intern@ubunto:~/Downloads$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl4
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl4
1 upgraded, 1 newly installed, 0 to remove and 472 not upgraded.
Need to get 194 kB/487 kB of archives.
After this operation, 460 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.30 [194 k
]
Fetched 194 kB in 1s (156 kB/s)
(Reading database ... 205703 files and directories currently installed.)
Preparing to unpack .../libcurl4_7.81.0-1ubuntu1.30_amd64.deb ...
Unpacking libcurl4:amd64 (7.81.0-1ubuntu1.30) over (7.81.0-1ubuntu1.17) ...
Selecting previously unselected package curl.
Preparing to unpack .../curl_7.81.0-1ubuntu1.30_amd64.deb ...
Unpacking curl (7.81.0-1ubuntu1.30) ...
Setting up libcurl4:amd64 (7.81.0-1ubuntu1.30) ...
Setting up curl (7.81.0-1ubuntu1.30) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...

Progress: [ 89%] [#################################################################################.........]
```

*Figure 17. Installing curl.*

## 5.2 Check DVWA response headers

```
curl -I http://127.0.0.1/DVWA/
```

Purpose: Sends an HTTP HEAD request to DVWA and displays response headers. The evidence shows HTTP/1.1 302 Found with Location: login.php. This is a normal redirect to the DVWA login page.

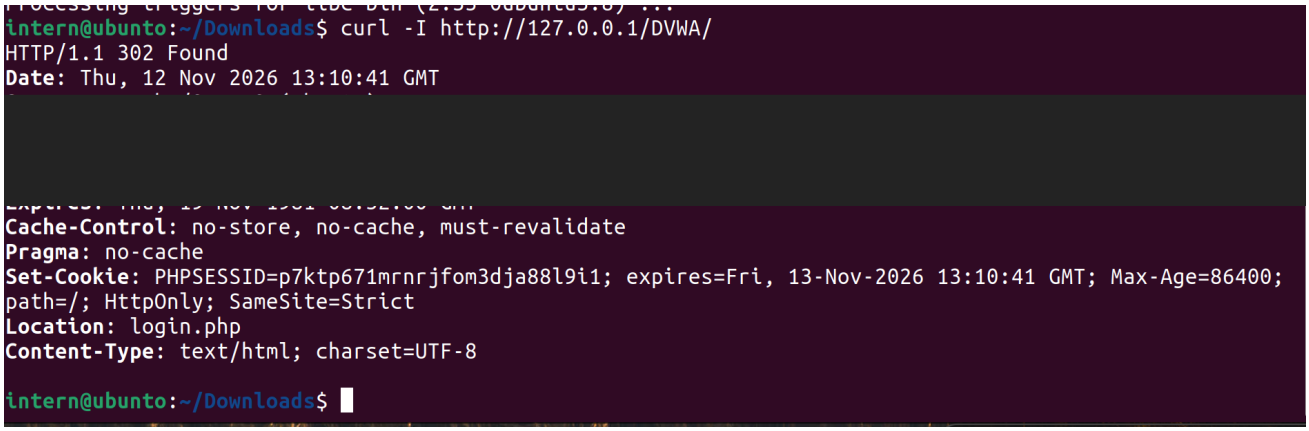



*Figure 18. DVWA HTTP response. Session identifiers have been redacted.*

### 5.3 Check the web server root response

```
curl -I http://127.0.0.1/
```

Purpose: Checks the HTTP headers returned by the Apache root page. The evidence shows HTTP/1.1 200 OK, confirming the root page is available.

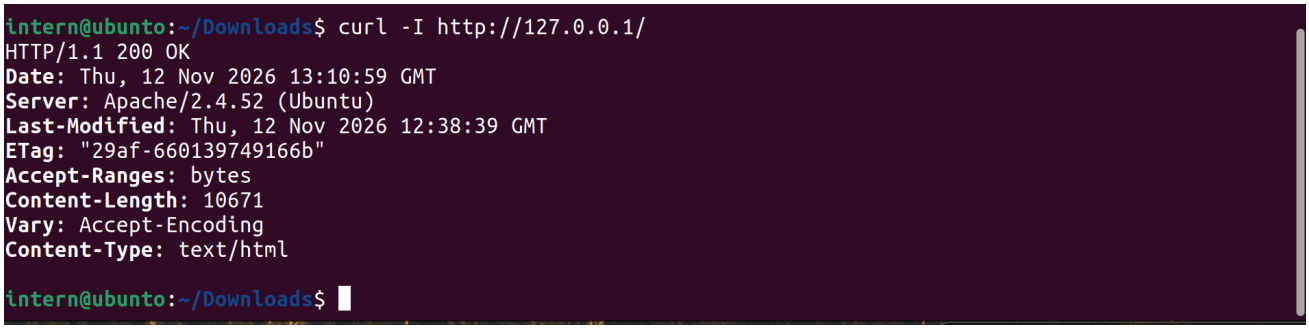



*Figure 19. Apache root page returns HTTP 200 OK.*

Supervisor clarification: http://172.16.229.131/ displays the Apache default page, while http://172.16.229.131/DVWA/ displays the DVWA login page. Both observations are consistent because they refer to different URL paths.

## 6. Gobuster Directory and File Enumeration

### 6.1 Install Gobuster

```
sudo apt install gobuster -y
```

Purpose: Installs Gobuster, a command-line tool for discovering web directories and files by requesting candidate paths from a wordlist.

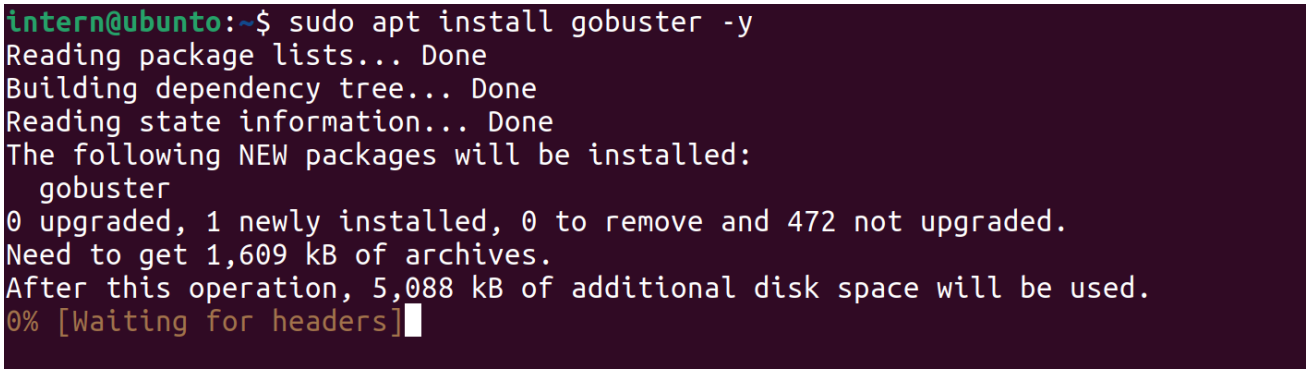



*Figure 20. Installing Gobuster.*

### 6.2 Install DIRB to obtain a wordlist

```
sudo apt install dirb -y
```

Purpose: DIRB was installed to obtain its bundled wordlists. Gobuster remains the enumeration tool used for the scan.

```
intern@ubunto:~$ sudo apt install dirb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  dirb
0 upgraded, 1 newly installed, 0 to remove and 472 not upgraded.
Need to get 203 kB of archives.
After this operation, 1,505 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu jammy/universe amd64 dirb amd64 2.22+dfsg-5 [203 kB]
Fetched 203 kB in 1s (137 kB/s)
Selecting previously unselected package dirb.
(Reading database ... 205715 files and directories currently installed.)
Preparing to unpack .../dirb_2.22+dfsg-5_amd64.deb ...
Unpacking dirb (2.22+dfsg-5) ...
Setting up dirb (2.22+dfsg-5) ...
Processing triggers for man-db (2.10.2-1) ...
```

*Figure 21. Installing DIRB to obtain its wordlist.*

## 6.3 Verify available wordlists

```
ls -lh /usr/share/dirb/wordlists/
```

Purpose: Lists the available wordlist files and their sizes. The scan used common.txt.

```
intern@ubunto:~$ ls -lh /usr/share/dirb/wordlists/
total 260K
-rw-r--r-- 1 root root 180K 2012  25  جنوری big.txt
-rw-r--r-- 1 root root 1.3K 2012  27  جنوری catala.txt
-rw-r--r-- 1 root root  36K 2014  17  نومبر common.txt
-rw-r--r-- 1 root root 1.5K 2012  23    مئی euskera.txt
-rw-r--r-- 1 root root  142 2005  29  سمبر extensions_common.txt
-rw-r--r-- 1 root root   75 2012  16   مارچ indexes.txt
-rw-r--r-- 1 root root  244 2005  29  سمبر mutations_common.txt
drwxr-xr-x 2 root root 4.0K 18:23 12  نومبر others
```

*Figure 22. Verifying the DIRB wordlist directory and common.txt.*

## 6.4 Run Gobuster against the DVWA path

```
gobuster -m dir -u http://127.0.0.1/DVWA/ -w /usr/share/dirb/wordlists/common.txt -x php,txt,html
```

Purpose: -m dir selects directory mode; -u sets the target URL; -w selects the wordlist; -x adds php, txt and html extensions to candidate names. The installed version shown in the evidence is Gobuster 2.0.1, which uses this legacy syntax (rather than the newer 'gobuster dir' syntax).

```
intern@ubunto:~$ gobuster -m dir -u http://127.0.0.1/DVWA/ -w /usr/share/dirb/wordlists/common.txt -x ph
p,txt,html

=====================================================
Gobuster v2.0.1              OJ Reeves (@TheColonial)
=====================================================
[+] Mode         : dir
[+] Url/Domain   : http://127.0.0.1/DVWA/
[+] Threads      : 10
[+] Wordlist     : /usr/share/dirb/wordlists/common.txt
[+] Status codes : 200,204,301,302,307,403
[+] Extensions   : php,txt,html
[+] Timeout      : 10s
=====================================================
2026/11/12 18:25:19 Starting gobuster
=====================================================
/.hta (Status: 403)
/.hta.php (Status: 403)
/.hta.txt (Status: 403)
/.hta.html (Status: 403)
/.htpasswd (Status: 403)
/.htpasswd.php (Status: 403)
/.htpasswd.txt (Status: 403)
/.htpasswd.html (Status: 403)
/.git/HEAD (Status: 200)
/.htaccess (Status: 403)
```

*Figure 23. Gobuster scan configuration and first discovered paths.*

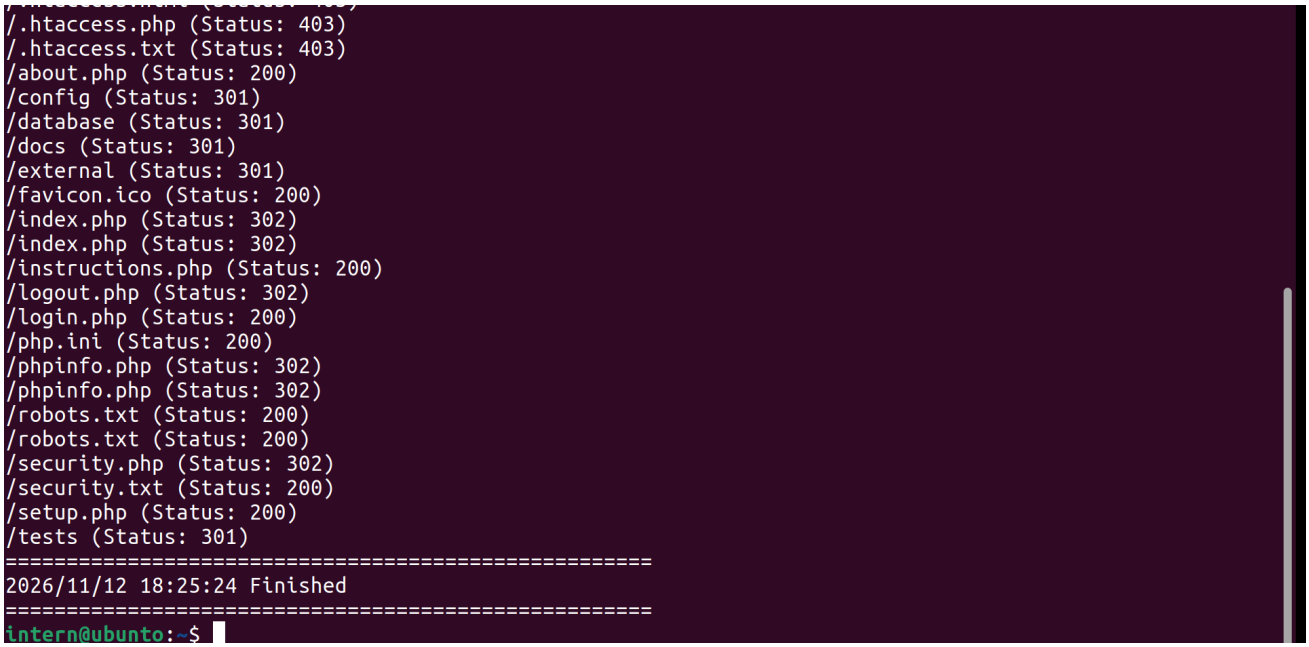



*Figure 24. Additional Gobuster discoveries and scan completion.*

## 6.5 Enumerated paths and HTTP status codes

| Path | Status | What the response indicates |
|---|---|---|
| /.git/HEAD | 200 | A response was returned; validate actual Git metadata exposure before declaring a finding. |
| /about.php | 200 | PHP page responded successfully. |
| /config | 301 | Redirect response; inspect destination. |
| /database | 301 | Redirect response; inspect destination. |
| /docs | 301 | Redirect response; inspect destination. |
| /external | 301 | Redirect response; inspect destination. |
| /favicon.ico | 200 | Favicon resource responded. |
| /index.php | 302 | Redirect response. |
| /instructions.php | 200 | Instructions page responded. |
| /logout.php | 302 | Redirect response. |
| /login.php | 200 | Login page responded. |
| /php.ini | 200 | Response returned; inspect content to determine whether configuration is exposed. |
| /phpinfo.php | 302 | Redirect response. |
| /robots.txt | 200 | Robots file responded. |
| /security.php | 302 | Redirect response. |
| /security.txt | 200 | Security text file responded. |
| /setup.php | 200 | DVWA setup page responded. |
| /tests | 301 | Redirect response; inspect destination. |

The scan also returned HTTP 403 for .hta, .htaccess and .htpasswd variants. A 403 response means the request was forbidden; it does not prove that the underlying file is readable.

Important interpretation: HTTP 200, 301, 302 and 403 are response codes, not vulnerability verdicts. In particular, /.git/HEAD and /php.ini returning 200 are leads requiring manual content validation. Redirects should be followed and inspected before making security claims.

# 7. Consolidated Results, Limitations and Conclusion

## 7.1 Initial attack surface summary

| Assessment area | Result |
|---|---|
| Host discovery | VM 172.16.229.131 was reachable. |
| Network service | TCP port 80 open; HTTP service identified. |
| Web server | Apache 2.4.52 on Ubuntu. |
| Application | DVWA accessible at /DVWA/. |
| Nmap root title | Apache default page at /. |
| Directory enumeration | Gobuster 2.0.1 discovered multiple application paths and files. |
| Potential items for validation | /.git/HEAD and /php.ini returned 200; several paths returned redirects. |

## 7.2 Limitations

This is an initial reconnaissance report, not a complete vulnerability assessment. No exploit validation, CVSS scoring, hardening or retest is claimed in this document. Gobuster's results are based on the common wordlist and the configured extensions, so undiscovered paths may still exist. Nmap's OS fingerprint is approximate. Some results require manual inspection.

## 7.3 Conclusion

The lab target was prepared with Apache, MariaDB, PHP and DVWA. Nmap identified Apache HTTP on TCP port 80 at 172.16.229.131. curl confirmed that the root path returns HTTP 200 and the DVWA path redirects to its login page with HTTP 302. Gobuster 2.0.1 enumerated multiple paths under /DVWA/, including pages, text files, redirects and forbidden paths. These results establish the initial attack-surface baseline for subsequent testing.

Safety: Keep DVWA isolated and use it only for authorized training. Do not deploy this intentionally vulnerable application on a public-facing production server.

# Appendix A – Command Reference

| Task | Command |
|---|---|
| Update packages | sudo apt update |
| Install web/database/PHP stack | sudo apt install apache2 mariadb-server php libapache2-mod-php php-mysql php-gd php-curl php-mbstring php-xml php-zip git unzip -y |
| Enable services | sudo systemctl enable apache2 && sudo systemctl enable mariadb |
| Start services | sudo systemctl start apache2 && sudo systemctl start mariadb |
| Check services | sudo systemctl status apache2; sudo systemctl status mariadb |
| Verify versions | apache2 -v; php -v; mysql --version |
| Clone DVWA | sudo git clone https://github.com/digininja/DVWA.git /var/www/html/DVWA |
| Set ownership | sudo chown -R www-data:www-data /var/www/html/DVWA |

| Task | Command |
|---|---|
| Copy config template | sudo cp config.inc.php.dist config.inc.php |
| Create DB/user | CREATE DATABASE dvwa; CREATE USER ...; GRANT ALL PRIVILEGES ...; FLUSH PRIVILEGES; |
| Find VM IP | ip addr |
| Install Nmap | sudo apt install nmap -y |
| Verify Nmap | nmap --version |
| Nmap scan | sudo nmap -sV -A 172.16.229.131 -oN reconnaissance/nmap/initial-scan.txt |
| Install curl | sudo apt install curl -y |
| Check DVWA headers | curl -I http://127.0.0.1/DVWA/ |
| Check root headers | curl -I http://127.0.0.1/ |
| Install Gobuster | sudo apt install gobuster -y |
| Install wordlists | sudo apt install dirb -y |
| List wordlists | ls -lh /usr/share/dirb/wordlists/ |
| Gobuster scan | gobuster -m dir -u http://127.0.0.1/DVWA/ -w /usr/share/dirb/wordlists/common.txt -x php,txt,html |

# Appendix B – Repository Evidence Map

| Repository path | Evidence |
|---|---|
| reconnaissance/nmap/initial-scan.txt | Raw Nmap scan output |
| reconnaissance/gobuster/dvwa-scan.txt | Raw Gobuster enumeration output |
| reconnaissance/initial-recon.md | Markdown reconnaissance summary |
| screenshots/01 dvwa-setup/ | DVWA installation and configuration screenshots |
| screenshots/02 nmap/ | IP, Nmap and curl screenshots |
| screenshots/03 directory-enumeration/ | Gobuster installation, wordlist and scan screenshots |
| lab/scripts/setup.sh | Lab setup shell script |
| README.md | Repository overview and usage information |

Prepared from the supplied screenshots and scan-output files. Commands and findings are transcribed from the available evidence; where an operation's exact terminal output was not visible in the supplied screenshot, the report describes the purpose without inventing an output.